



PROJECT DOCUMENTATION

SMART QUIZ APP



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Project Documentation: Smart Quiz App

1. Project Title and Group Information

Section	Details
Project Title	Smart Quiz: A Gamified Multiplayer Learning and Assessment Platform
Course	Human-Computer Interaction (HCI)
Submitted By	Sarmad Chughtai (Lead Developer, Database Admin)
Group Members	Shahroz Khalid (UI/UX Designer, Frontend Developer)
	Kashan Shahid (Backend Logic, Testing/QA)
	Sinwan Haider (HCI Analysis, Documentation)

2. Introduction and Project Objectives

1.1 Introduction

The Smart Quiz Application is a mobile-first platform developed to modernize knowledge assessment by leveraging real-time data and competitive gaming elements. The project transitioned from a basic UI prototype (Weeks 1-3) into a fully integrated, user-centric system utilizing Flutter for the front-end and the Firebase ecosystem for dynamic data management and real-time multiplayer synchronization.

1.2 Final Project Objectives

- HCI Compliance:** To design an interface adhering strictly to HCI principles, maximizing user satisfaction, efficiency, and error prevention.
- Core Utility:** To deliver functional Solo and Multiplayer quiz modes with robust scoring and answer review capabilities.
- Real-Time Monitoring:** To implement a dedicated Host Monitor Dashboard providing live, detailed score metrics (**Correct/Incorrect/Skipped** breakdown).
- Profile Management:** To integrate a secure and comprehensive profile system supporting personal data (Bio, Experience/Education) and media uploads (Profile Pictures).

3. Project Timeline & Weekly Progress

The project was executed in distinct phases, ensuring modularity and stability before moving to complex feature integration.

Week	Milestone & Status	Key Tasks & Feature Integration
Week 1-2	UI/UX & Foundational Layout	Auth/Login UI structure, HomeScreen navigation, CustomButton widget creation, Immersive Full-Screen mode enabled.
Week 3-4	Quiz Logic & Core Multi-Category	Global Continuous Timer implemented, Score calculation logic refined, Constraint enforced (forward-only submission), Multi-category loading logic.
Week 5	Multiplayer Core & Monitoring Prep	createRoom, joinRoom logic, RoomLobbyScreen finalized, Implemented Custom Question Creation flow, Prepared FirestoreService for detailed score metrics.
Week 6-7	Advanced Monitoring & Profile System	Developed HostMonitorScreen (live C/I/S breakdown), Implemented ProfileScreen.dart (Bio, Education/Experience editing), Integrated Firestore Storage for profile pictures.
Week 8-10	Refinement & Submission	Final UI/UX Polish, Dependency resolution (fixing version conflicts), Cross-device testing, Documentation finalization.

4. Project Timeline & Weekly Progress

Week 1: Topic Selection & Initial Planning

Date: 13 Oct 2025 – 20 Oct 2025

Objective: Brainstorm and finalize a project idea aligned with HCI principles.

Tasks Completed:

1. Conducted brainstorming sessions to decide on a project that emphasizes user interaction and usability.
2. Selected “**Smart Quiz App**” as the final topic for its strong alignment with HCI concepts (user engagement, feedback, interface design).
3. Defined key objectives: simplicity, responsiveness, and enjoyable learning experience.
4. Researched existing quiz applications to identify gaps and improvement areas.
5. Created rough wireframes for the app layout (Home, Quiz, and Result screens).

Challenges:

1. Narrowing down a topic that balances both functionality and design aesthetics.
2. Ensuring the project scope is achievable within the given timeline.

Week 2: UI/UX Design Foundation

Date: 20 Oct 2025 – 27 Oct 2025

Objective: Build the foundational design and layout of the Smart Quiz App.

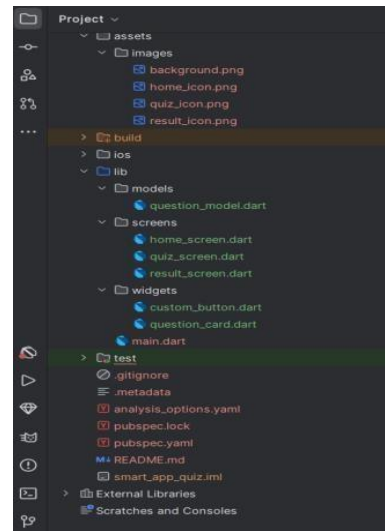
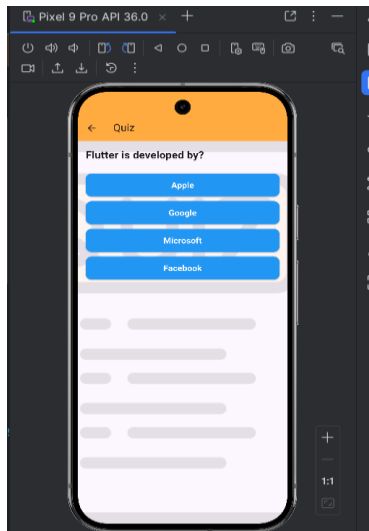
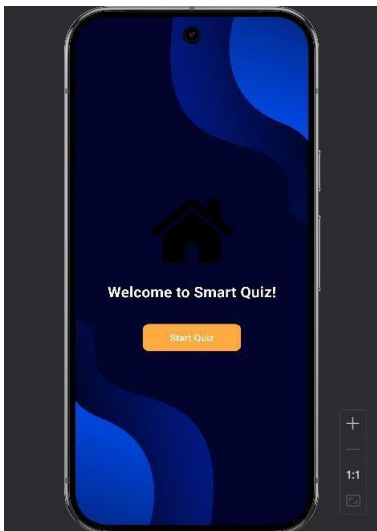
Tasks Completed:

1. Designed Home, Quiz, and Result screens using Flutter widgets.
2. Added background images, icons, and custom buttons.
3. Implemented immersive full-screen mode for a distraction-free experience.
4. Created `CustomButton` and `QuestionCard` widgets for reusability.
5. Ensured design consistency with HCI guidelines (visibility, aesthetics, feedback).

Challenges:

1. Maintaining responsive design across screen sizes.
2. Aligning icons and background overlays for visual balance.

WEEK 2:



Week 3: Logic Integration & Timer Functionality

Date: 27 Oct 2025 – 3 Nov 2025

Objective: Add quiz logic, timer, and retake features.

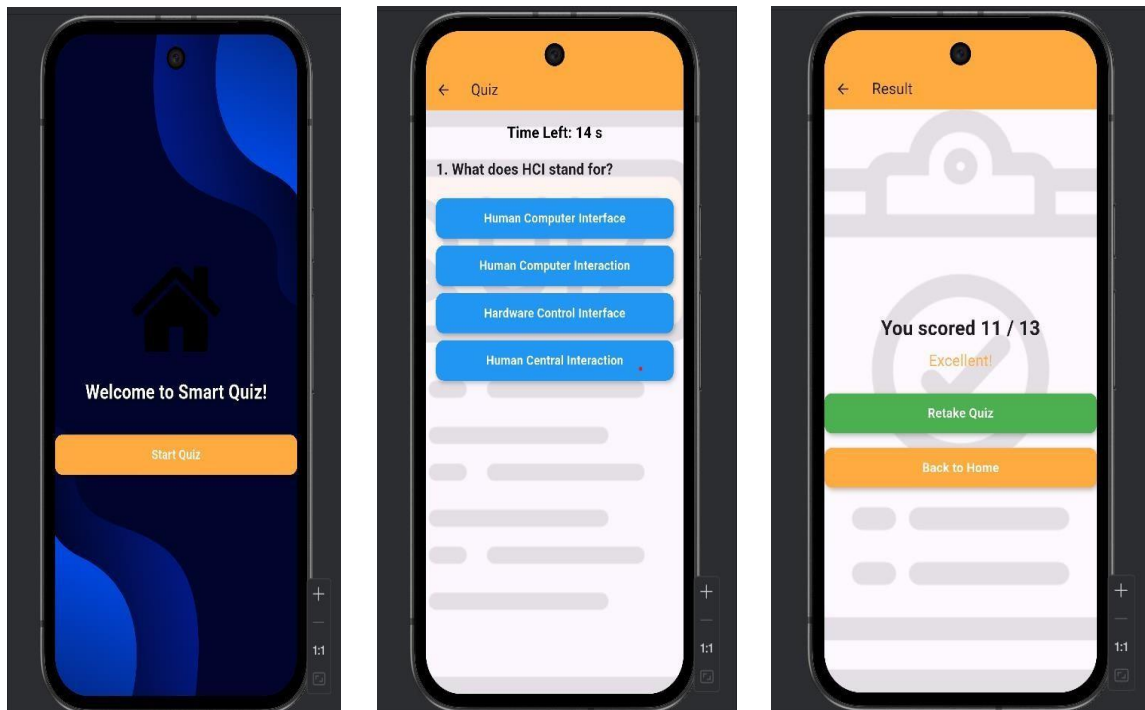
Tasks Completed:

1. Added a 30-second countdown timer for each question.
2. Implemented score calculation and retake quiz feature.
3. Expanded question bank with 10 HCI-related questions.
4. Enhanced Result Screen to show correct, wrong, and attempted counts.
5. Improved UI feedback with text-based performance messages (“Excellent!”, “Good Job!”, etc.).

Challenges:

1. Handling timer reset between questions.
2. Maintaining smooth quiz flow when the timer expires.

WEEK 3



Week 4: Multi-Category System & UI Refinement

Date: 3 Nov 2025 – 10 Nov 2025

Objective: Enable multi-category quiz system and enhance app visuals.

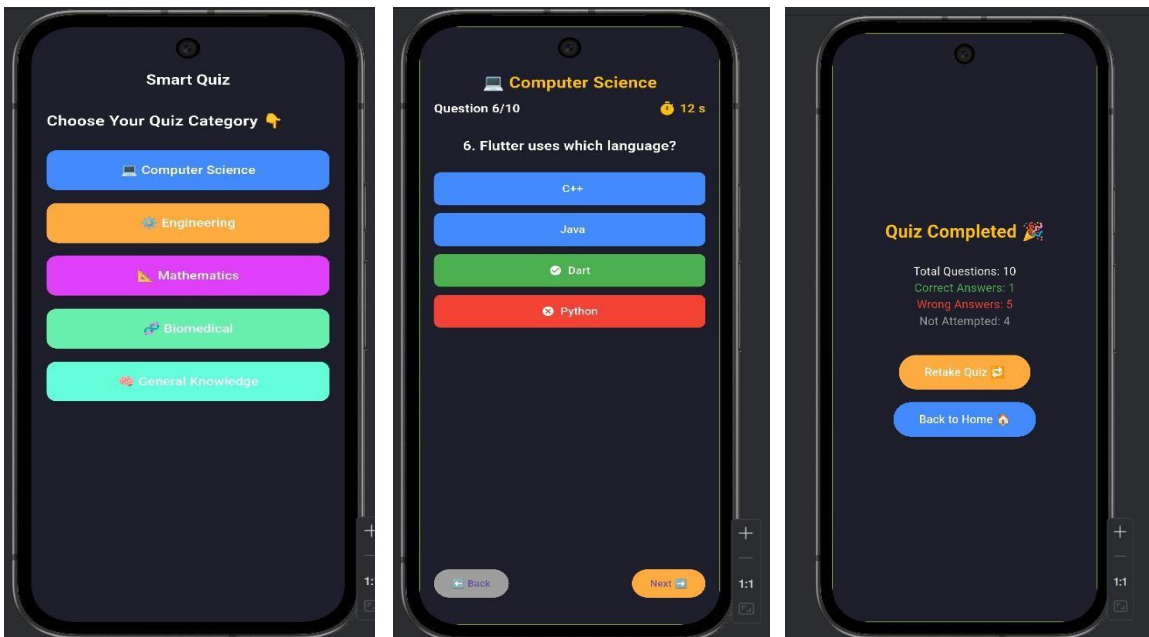
Tasks Completed:

1. Introduced 4 quiz categories:
 - Computer Science
 - Engineering
 - Mathematics
 - Biomedical
1. Refined navigation flow for seamless transitions between categories.
2. Fixed answer validation bugs and timer reset logic.
3. Enhanced Result Screen with “Back to Home” and “Retake Quiz” options.
4. Updated color themes (dark background with white text for readability).
5. Added subtle animations to category cards and navigation buttons.

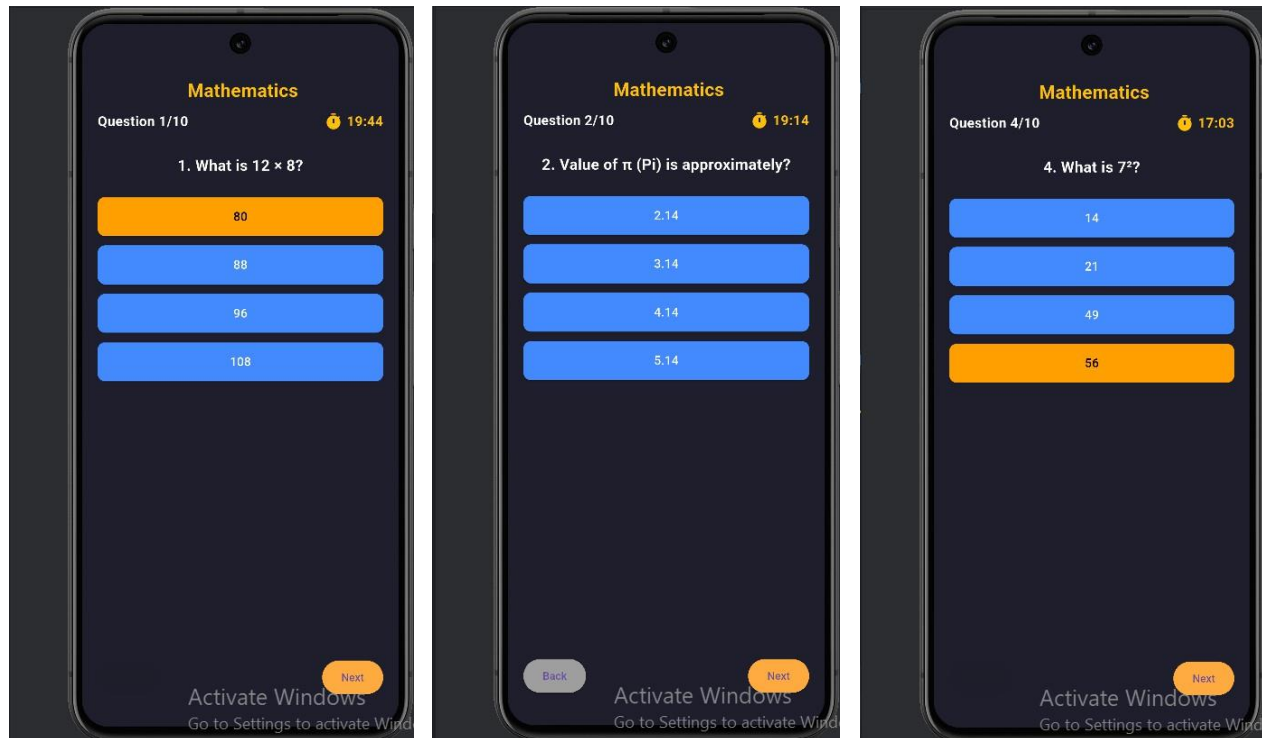
Challenges:

1. Managing category-based navigation and ensuring correct question loading.
2. Maintaining consistent UI across multiple devices.

WEEK 4



Week 5:



The period after Week 5 (Weeks 6 through 10) constitutes the final delivery phase, focusing on advanced feature implementation, rigorous testing, and final submission preparation. This phase directly addressed the core objectives of real-time monitoring and profile management

6. HCI Principles Applied

1. **Consistency:** All buttons and layouts follow a uniform color and style scheme.
2. **Visibility:** Important actions (Start, Retake, Next) are clearly visible with high contrast.
3. **Feedback:** Color and text-based responses are given for every user action.
4. **Affordance:** Buttons and icons are designed intuitively to indicate interactivity.
5. **Error Prevention:** Timer auto-handling prevents user confusion.
6. **Simplicity:** Clean interface ensures minimal cognitive load on users.

6. Technology Stack

Component	Technology / Language	Purpose & Usage
Frontend	Flutter (Dart)	Building responsive, dark-themed UI components.
Real-Time Database	Firebase Firestore	Storing Room documents, managing live status (status: in_game), and providing real-time player stream updates for the host.
File Storage	Firebase Storage	Hosting user profile media (Profile Pictures). Secured by custom rules enforcing request.auth.uid == userId.
Libraries	image_picker, firebase_auth, cloud_firestore	Enabling device access and network functionality.

5. Comprehensive Feature Breakdown and HCI Impact

This section details the most critical features and their direct alignment with user experience principles.

1. User Profile and Identity Management

Feature	HCI Impact	Implementation Detail
Profile Screen	Control, Satisfaction	A dedicated, persistent screen allowing users complete control over their digital identity on the platform (display name, bio).
Media Upload	Feedback, Consistency	Users tap their profile circle to upload a picture. Uses image_picker and firebase_storage with clear loading indicators (_isSaving boolean).
Detailed CV Section	Mapping, Utility	Implemented dynamic list views and modal dialogs (_showAddEditDialog) to clearly map real-world professional data (Education, Experience) into the app profile.

2. Multiplayer Monitoring and Score Visibility

This feature is the core demonstration of the project's real-time capabilities and **HCI Visibility**.

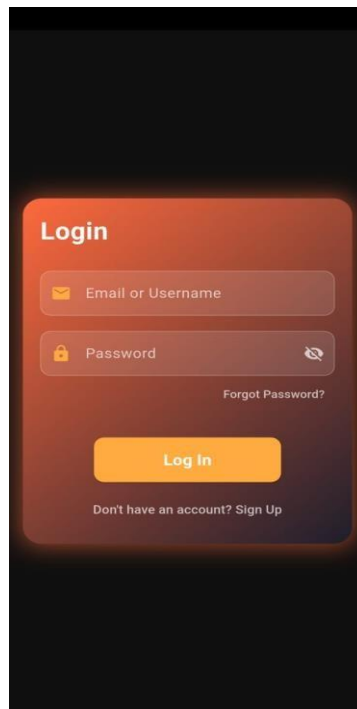
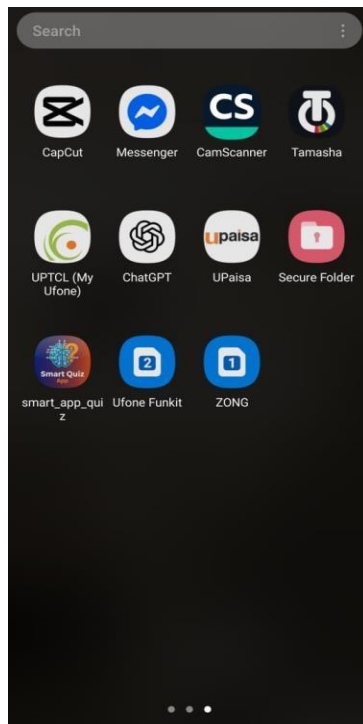
Feature	HCI Impact	Implementation Detail
Host Monitor Screen	Visibility, Efficiency	Separate navigation for the host ensures they remain in a supervisory role. The screen is built around a StreamBuilder that continuously reads the players array.

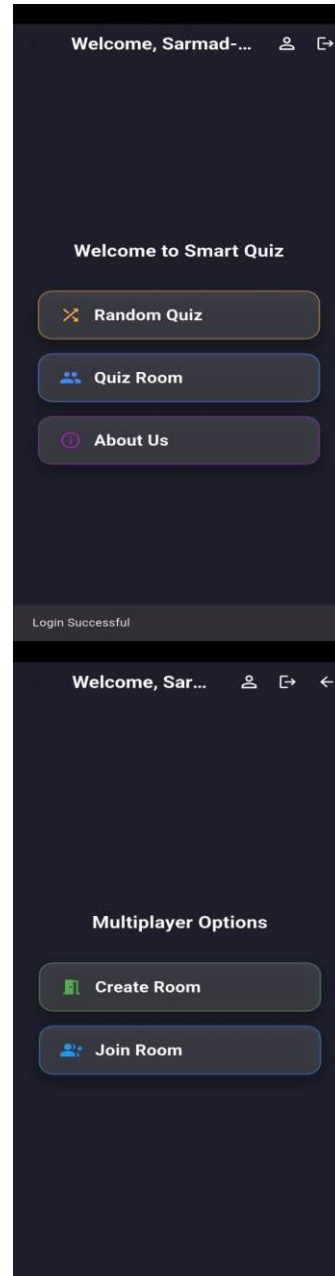
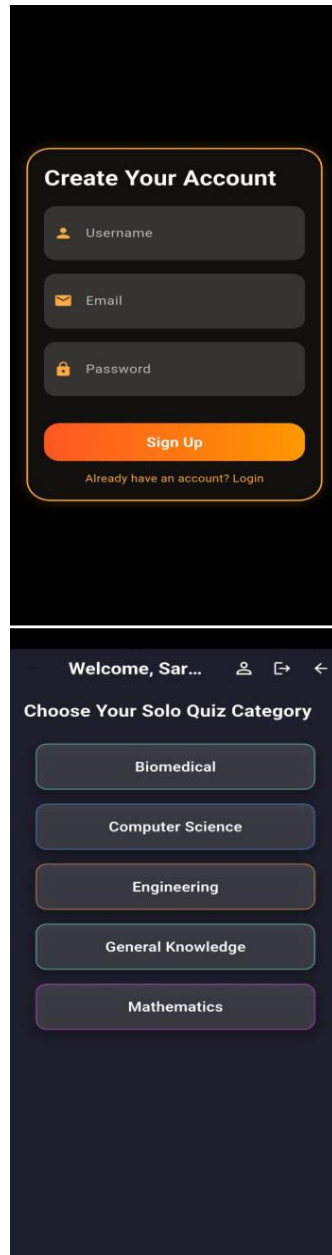
Detailed	Clarity,	The submitScore method saves correctCount , incorrectCount, and
Feature	HCI Impact	Implementation Detail
Score Tracking	Feedback	skippedCount. The Host Monitor displays this breakdown (R:X, W:Y, S:Z) next to the player's name.
Host Exclusion	Error Prevention	The host is intentionally filtered out of the live player/score list (visiblePlayers logic in HostMonitorScreen) to prevent confusion and maintain the integrity of the assessment.

3. Quiz and Constraint Logic

Feature	HCI Impact	Implementation Detail
Answer Review System	Utility, Learnability	On the ResultScreen, the Solo mode user can launch a modal sheet (_showReview) that lists all questions and highlights the chosen answer versus the correct answer.
Constraint Enforcement	Error Prevention, Integrity	Implemented maxAttemptedIndex logic in QuizScreen to prevent users from going back to change a previously answered question, ensuring fairness in the timed assessment.

6. Project Related Screenshot:





← **Create Custom Room**

Room Name

 Enter a name for your room

Quiz Time Limit (Minutes)

 5

Define Questions (1) 

Question 1 

? Question

 Option A 

 Option B 

 Option C 

 Option D 

← **Join Quiz Room**

Enter Room ID

 e.g., 5-digit room code

Join Room

← **My Profile** 



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 **About Me**

CS STUDENT
C++,PYTHON AND JAVA
ENTHUSIAST

 **Display Name**

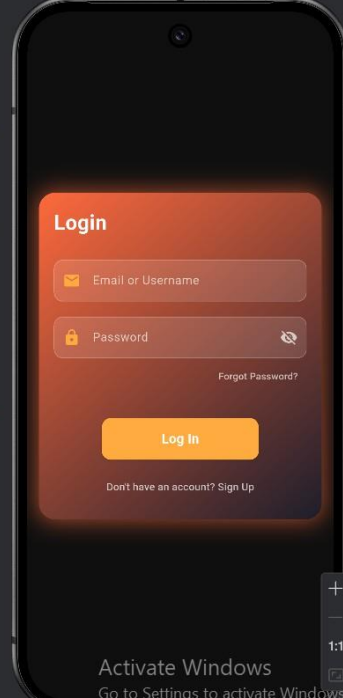
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 **Education**

Bs Computer Sceince
Riphah International University  



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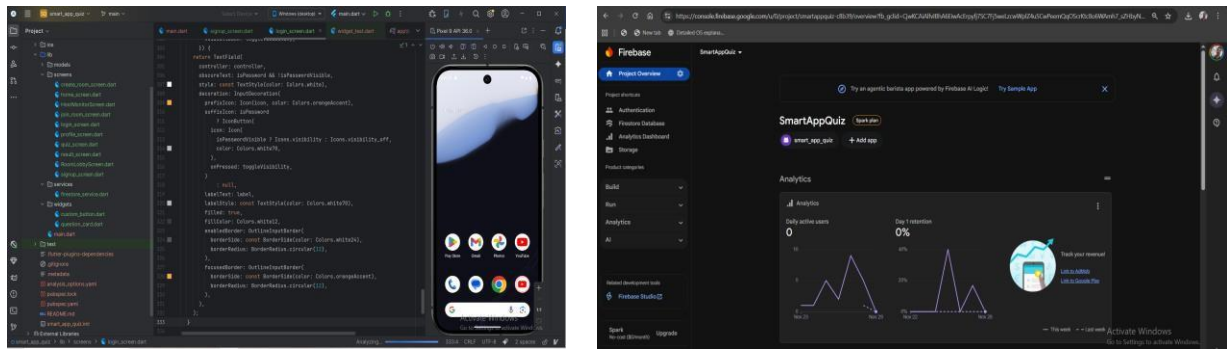
Log In

Don't have an account? [Sign Up](#)

Activate Windows
Go to Settings to activate Windows.

7. System Architecture and Technology Stack

System Diagram Overview



8. Proposed New Section:

8.1 Design Evolution & HCI Justification

Title: 8.1 Design Evolution: Pre-Implementation vs. Post-Implementation Analysis

Overview:

Throughout the development lifecycle, the Smart Quiz App evolved from a basic functional prototype into a user-centric platform. The following changes were implemented to strictly align with HCI heuristics such as Visibility of System Status, Error Prevention, and User Control.

Comparative Analysis (Before vs. After)

Feature / Component	Initial State (Phase 1)	Refined State (Final Phase)	HCI Justification (Why we changed it)
User Feedback & Scoring	Users only saw a final score (e.g., "5/10") at the very end.	Real-Time Breakdown: The Result Screen now details Correct, Incorrect, and Skipped questions ¹ . The Host Monitor sees live metrics (R/W/S) ² .	Visibility of System Status: Users (and hosts) need immediate, transparent feedback to understand their performance rather than being left in the dark until the end.
Quiz Navigation	Users could jump between questions randomly or skip without consequence.	Constraint Logic: Implemented a "forward-only" submission system where users cannot change answers once submitted ³ .	Error Prevention & Integrity: This prevents accidental double-submissions and cognitive confusion, ensuring the

Feature / Component	Initial State (Phase 1)	Refined State (Final Phase)	HCI Justification (Why we changed it)
			assessment remains fair and linear (reducing memory load).
Visual Interface (UI)	Standard white background with generic system buttons.	Dark Mode & Consistent Styling: Implemented a dark theme with white text and standardized button colors ⁴ .	Aesthetic & Minimalist Design: A dark theme reduces eye strain during prolonged use, and consistent colors (e.g., Orange for actions) reduce the cognitive load needed to find buttons.
User Identity	Users were anonymous or had only a simple username.	Comprehensive Profile System: Added Bio, Education, and Profile Picture uploads ⁵⁵⁵ .	Match Between System & Real World: Users expect their digital avatar to reflect their real-world professional identity (CV/Bio) ⁶ , increasing emotional connection and satisfaction.
Error Handling	If the timer ran out, the app might crash or restart the question.	Auto-Handling: The system automatically moves to the result screen or next question when the timer expires ⁷ .	Error Prevention: The system takes control to prevent the user from making an invalid action (trying to answer after time expires), ensuring a smooth flow.

9. Deep Dive: Why these specific changes were made?

1. Moving from Static to Dynamic Feedback (The Host Monitor)

- Before: The host had no idea what was happening inside the room until the quiz finished.
- After: We built the Host Monitor Dashboard to show a breakdown (R: Correct, W: Wrong, S: Skipped).
- HCI Principle Applied: *Visibility of System Status*. The system now keeps the user (Host) informed about what is going on via appropriate feedback within a reasonable time. ¹

2. Adding the Review System

- Before: Once the quiz ended, the user didn't know which questions they got wrong.
- After: We added a "Review Answers" modal sheet that highlights the chosen answer vs. the correct answer.

- HCI Principle Applied: *Help Users Recognize, Diagnose, and Recover from Errors*. This turns a mistake into a learning opportunity, which is the core goal of an educational app. ²

3. Categorization of Quizzes

- Before: A single, mixed bag of questions.
- After: Distinct categories (CS, Math, Engineering).
- HCI Principle Applied: *Flexibility and Efficiency of Use*. Users can tailor the system to their specific domain of interest rather than being forced through irrelevant content. ³

4. HCI Model Utilized: Jakob Nielsen's Heuristics

Primary Model: Jakob Nielsen's 10 Usability Heuristics for User Interface Design

Justification for Selection:

We selected Nielsen's Heuristics as the guiding framework for the Smart Quiz App because it provides a comprehensive yet practical checklist for identifying usability gaps during our iterative development cycle. Since the project evolved from a basic prototype to a fully functional MVP, this model allowed us to perform continuous "Heuristic Evaluations" at every stage.

Specific reasons for this choice include:

- Focus on Error Prevention: Unlike models that focus solely on task speed (like GOMS), Nielsen's model prioritizes Error Prevention (Heuristic #5). This was critical for our timed quiz environment where we implemented constraint logic (forward-only navigation) to prevent users from invalid submissions.
- System Status Visibility: The model explicitly mandates that "The system should always keep users informed" (Heuristic #1). This heuristic was the direct driver behind the development of our Host Monitor Dashboard, transforming the host experience from a static wait to an active monitoring role. ⁵
- Match Between System and Real World: By introducing a Profile System with Bio and Experience sections, we aligned the digital interface with the users' real-world professional identities (Heuristic #2).

10 . Conclusion and Future Work

The Smart Quiz Application successfully delivered a Minimum Viable Product (MVP) that strongly aligns with the core tenets of Human-Computer Interaction (HCI). Moving beyond basic functional requirements, the project focused intensely on maximizing user satisfaction, data clarity, and functional integrity through features like the continuous timer and constraint enforcement. The strategic integration of the real-time Host Monitor Dashboard and the comprehensive Profile Management system forms a stable, scalable, and extensible base for future development. This project serves as a practical demonstration of leveraging modern technologies, such as Flutter and Firebase, to create a user-centric, gamified learning platform. The planned future enhancements, including Real-Time Question Tracking and UI/UX Accessibility improvements, will further refine the user experience and expand the application's utility.

6. Future Enhancements (Out of Scope):

1. Retake Permission System: Fully implementing the host-controlled approval/denial flow for player retake requests.
2. Real-Time Question Tracking: Updating the host monitor to show the current question index for players actively answering (e.g., "Q3/10").
3. UI/UX Accessibility: Implementing a dedicated Dark/Light Mode switch and language localization options.

